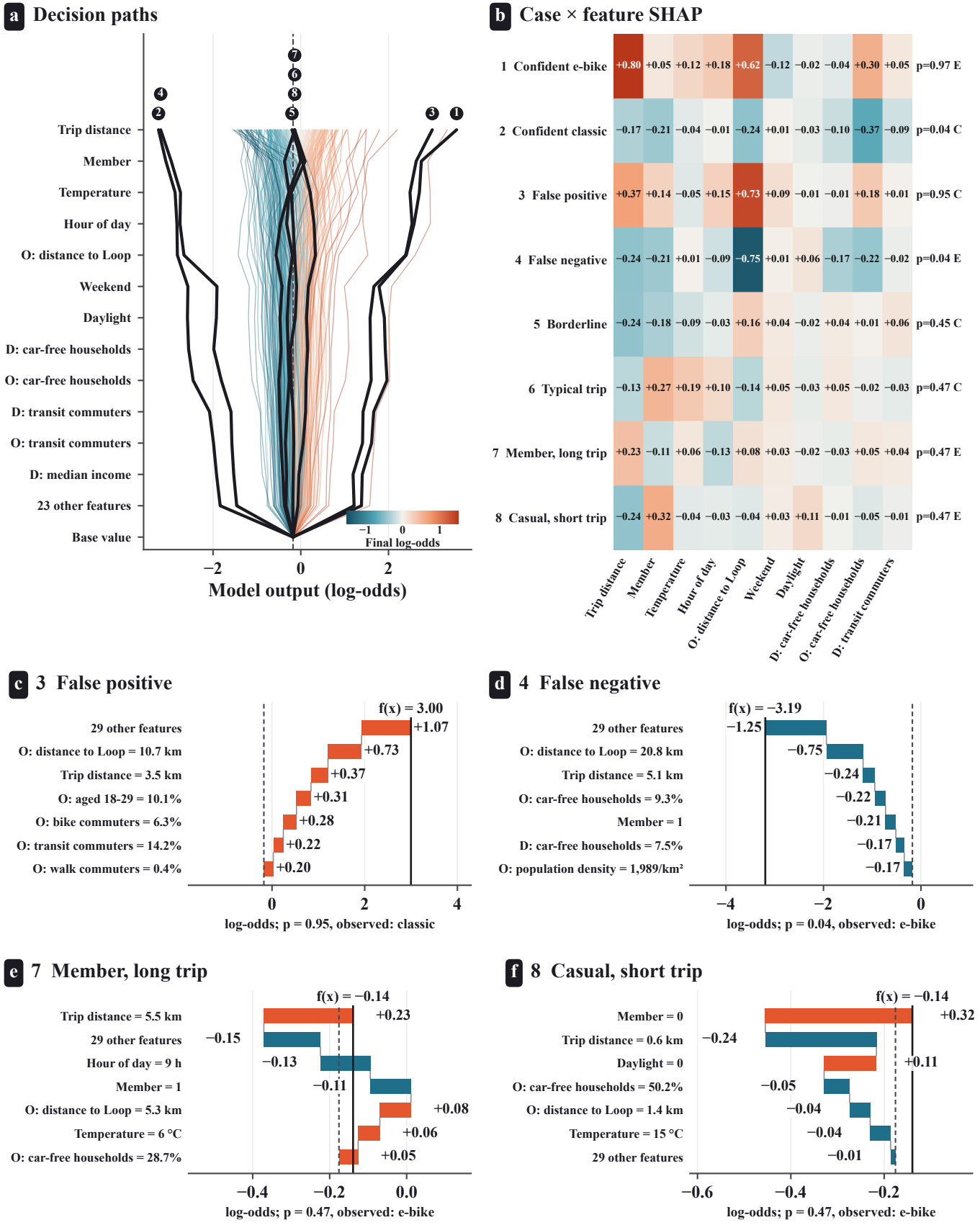

a Decision paths
Trip distance
Member
Temperature
Hour of day
O: distance to Loop
Weekend
Daylight
D: car-free households
O: car-free households
D: transit commuters
O: transit commuters
D: median income
23 other features
Base value
Final log-odds
Model output (log-odds)
b Case × feature SHAP
1 Confident e-bike
2 Confident classic
3 False positive
4 False negative
5 Borderline
6 Typical trip
7 Member, long trip
8 Casual, short trip
c 3 False positive
f(x) = 3.00
29 other features
O: distance to Loop = 10.7 km
Trip distance = 3.5 km
O: aged 18-29 = 10.1%
O: bike commuters = 6.3%
O: transit commuters = 14.2%
O: walk commuters = 0.4%
log-odds; p = 0.95, observed: classic
d 4 False negative
f(x) = −3.19
29 other features
O: distance to Loop = 20.8 km
Trip distance = 5.1 km
O: car-free households = 9.3%
Member = 1
D: car-free households = 7.5%
O: population density = 1,989/km²
log-odds; p = 0.04, observed: e-bike
e 7 Member, long trip
f(x) = −0.14
Trip distance = 5.5 km
29 other features
Hour of day = 9 h
Member = 1
O: distance to Loop = 5.3 km
Temperature = 6 °C
O: car-free households = 28.7%
log-odds; p = 0.47, observed: e-bike
f 8 Casual, short trip
f(x) = −0.14
Member = 0
Trip distance = 0.6 km
Daylight = 0
O: car-free households = 50.2%
O: distance to Loop = 1.4 km
Temperature = 15 °C
29 other features
log-odds; p = 0.47, observed: e-bike